

Management

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Deciphering Spotify : Strategic Behavioural Decisions Making in the Landscape of Digital Music

Abstract:

This paper is going to examine the strategic behavioural decision making process of Spotify in the digital music industry by focusing on its innovative use of advanced technologies. This paper also identifies certain cognitive biases that are present in Spotify with the help of case studies and understands how it affects user experience. Furthermore this paper explores the role of emerging technologies for enhancing user experience. This paper also provides actionable insights for other platforms on how they can use advanced data analytics and behavioural insights to improve the decision making for achieving sustainable growth.

Introduction:

In this era of digital transformation online platforms act as an intermediary between technology and customer engagement across various domains. Various platforms like Amazon, Facebook, Google, Spotify, YouTube, etc are harnessing the dynamic transformation of the global economy in this era of global platformisation through value creation on a global scale (Kenney & Zysman, 2020). I have chosen Spotify as my focus for this assignment as It is a global leader in music streaming. It completely changed the way in which songs and podcasts were consumed by the platform users by handling the production and distribution of music in a unique manner.

This study primarily focuses on Spotify which is an ideal example of a user-generated content in the digital music industry. By developing a platform where consumers not only listen to music and podcasts produced by other artists but also create and share their own content, Spotify has revolutionised the digitalisation of the music industry (Aguiar & Waldfogel, 2018). Spotify has democratised music distribution by giving independent artists a fair chance to showcase their talent by bypassing traditional music industry barriers such as record labels and distributors, providing them direct access to the global audiences (Morgan, 2019). Spotify has always utilised reviews and feedback from its user base to continuously refine and enhance its services (Wang, 2023).

Spotify's entry into the market has disrupted music distribution, affecting everything from digital piracy rates to how local artists monetize their work, illustrating the global significance of the platform (Aguilar & Waldfogel, 2021). Spotify has been facing significant competition from its rivals such as Apple Music, Amazon Music, Pandora, etc in the online music industry (Hracs & Webster, 2020).

This has established Spotify as a key contributor in the transformation of the online music industry, making it an ideal case study for examining strategic behavioural decision making in digital platforms. By integrating theoretical concepts with qualitative and quantitative analysis this study will cover several critical areas like impact of cognitive biases and heuristics on managerial judgments. It will critically examine how the integration of user behaviour analysis, technological innovations and adoption of cutting edge technologies can support Spotify's goals and enhance decision making efficiency to drive sustainable growth . Additionally this study will give actionable insights that could benefit other digital platforms to optimise their strategies in various sectors.

Platform Overview:

Digitalization has shifted traditional media industry replacing physical music carriers with music streaming platforms like Spotify (Lozić, Vojkovic, & Milković, 2020). Since its foundation in 2006, Spotify revolutionised the music industry by changing physical song purchases model to a subscription based streaming service model. This offers users unlimited access to a vast library of tracks, genres, albums, and artist collection. This not only changed how people consume music but also how artists distribute it and how labels profit from it. Spotify offers two kind of options a free advertising supported and a paid subscription based model. This influenced similar shifts across other media platforms like Apple Music, Amazon Music, Netflix, YouTube, etc changing the landscape of media consumption (Fleischer, 2020).

Features that distinguish Spotify from others include use of ML Models, Echo Nest Integration and User Behaviour Analysis to personalised playlists like Discover Weekly and Release Radar (Germain & Chakareski, 2013). ML algorithms used by Spotify include Matrix Factorization to capture user preferences and item characteristics, deep learning to extract

patterns that simpler models might miss to capture changes in user preferences over time, CNNs to detect patterns related to genre and rhythm and RNNs for features like auto play. It also uses GBMs to predict user ratings for songs or how likely a user is to skip a song after a few seconds, BNs to dynamically adapt recommendations based on changing user behaviour and reinforcement learning to learn from user interactions and quickly adapt to integrate feedback by providing recommendations as per their feedback.

Spotify also uses sentiment analysis to understand user emotions from reviews and comments and make recommendations more personalised as per user behaviour (Amrutha & Supriya, 2023). Spotify also expanded its content beyond music and radio to offer a platform for podcasts by acquiring podcast networks like Gimlet Media, Anchor, and Parcast which diversified its content offerings and expanded its audience reach (Prey, Valle, & Zwerwer, 2020). Spotify made music listening more accessible to everyone, making it easier and more affordable for users to access music anywhere and anytime. It has also provided emerging artists an opportunity to establish themselves as stars. This makes Spotify an influential platform to reach global audiences bypassing traditional music distribution channels. Due to these reasons Spotify continuously shapes the evolution of the music streaming industry with its innovations and user engagement.

Cognition and Heuristics Impact on Managerial Judgment:

1. Loss Aversion Bias:

Loss aversion bias refers to the scenario where the pain of losing is psychologically more significant than the pleasure of gaining something of equivalent value (Liu, 2023). This bias can influence human behaviour particularly during decisions that involve the risk of losing an existing advantage.

Case Study on Spotify: Subscription Cancellation

1. Spotify users face loss aversion bias while considering subscription cancellation (Bhartyadhikara, 2020).

2. Users prefer to keep premium features like advertisement free listening, mobile streaming, and high quality audio even if they don't use the service extensively.
3. **Observation:** It has been observed that users continue to pay for the subscription even if they do not use the service to its full extent, as the thought of losing premium benefits outweighs the logical decision to save money by cancelling (Redondo & Serrano, 2023).
4. **Outcome:** The users decide to keep the subscription, influenced by the desire to avoid the loss of premium features despite not using the service extensively (Bhartyadhikara, 2020).

2. Choice Overload Bias:

Choice overload occurs when the presence of too many options leads to confusion, anxiety and difficulty making a decision, which can result in dissatisfaction with the choice made (Chernev, Böckenholt & Goodman, 2015).

Case Study: Overwhelmed by Options

1. With millions of tracks available, Spotify users can experience choice overload, finding it challenging to decide what to listen to next (Chiravirakul & Payne, 2014).
2. These suggestions simplify choices by offering limited options tailored to the previous listening habits of the user.
3. **Observation and Outcome:** It is observed that despite having many choices, users often listen to a relatively narrow selection of music, showing how choice overload bias can lead users to rely on automatically tailored playlists like Release Radar, Daily Mix or Discover Weekly (Kamehkhosh, Bonnin & Jannach, 2019).

3. Popularity Bias (Superstar Effect):

Popularity bias occurs when individuals prefer items, products, or services that are popular among a large group (Turnbull et al., 2022). This bias can lead individuals to overlook less known options in favour of those that are highly rated.

Case Study: Promotion of Chart-Topping Artists

1. Algorithms in Spotify tend to promote songs that are already popular or trending (Trainor & Turnbull, 2023).
2. This creates a feedback loop where popular tracks become even more dominant, get more exposure, streamed by maximum people and hence rise higher in playlist.
3. **Observation :** It has been observed that when an independent artist releases a song it starts getting some attention at first. However, without algorithmic support to boost its visibility similar to top songs in the list it fails to reach a broader audience, despite some favourable reviews (Trainor & Turnbull, 2023).
4. **Outcome:** The song eventually becomes unknown, showing that Spotify's focus on popular songs can make it hard for new or independent artists to get noticed.

4. Framing Effect:

The framing effect describes how people make decisions based on how information is presented to them rather than considering the entire content (Martínez-Tur et al., 2018).

Case Study on Spotify: Subscription Options

1. Spotify presents its subscription options in a way that frames the premium service as vastly superior to the free version by highlighting benefits such as no advertisements, downloading offline music, playing songs in any order, high audio quality, listening with friends in real time and unlimited skips can make these features seem more valuable (Krisnawati & Aprilia, 2023).
2. **Observation:** Users might perceive the premium option as more valuable when its benefits are emphasized over the limitations of the free version (Suharyono & Astuti, 2020).
3. **Outcome:** This bias can influence users by framing the premium benefits in a positive manner to take the premium service subscription even if the free version meets their needs.

5. Anchor Bias

Anchor bias occurs when individuals rely too heavily on an initial piece of information (also known as the anchor) while making decisions (Song, Yang & Zhang, 2019).

Case Study: Early Ratings Influence

1. Initial ratings or reviews can serve as an anchor for shaping user perceptions of a track quality (Zhang, Zhou & Yang, 2012).
2. A few early high ratings can lead subsequent listeners to rate the song favourably, regardless of its actual merit potentially biasing overall ratings.
3. **Observation:** It has been observed that when subsequent listeners see the initial high ratings of the song and listen to the song with a positive mindset rate the song similarly high (Morris & Powers, 2015).
4. **Outcome:** The song receives a higher rating compared to that it might have received in a normal scenario, showing how initial opinions can influence subsequent user ratings.

Data Analysis and Decision Making Process:

1. Music Personalization at Spotify:

Approach:

According to (Jacobson et al., 2016) for making personalised playlist recommendations like Discover Weekly, Release Radar or Fresh Finds Spotify uses a combination of NLP, collaborative filtering and ML. Further it performs a cluster analysis on historical and current listening patterns to understand user preferences. To make these recommendations it collects data from two billion user generated playlists and monitoring daily activity of users.

Results:

Playlists curated based on Spotify recommendation through collaborative filtering and matrix factorization like Discover Weekly have reached one billion streams in 10 weeks. It was found that customising playlist as per consumer past choices and current listening patterns significantly improves user engagement and satisfaction. But the main challenge faced was that

Spotify only recommends familiar songs so users might miss out on listening to new music but if it focuses too much on new content users may not enjoy it as they are emotionally linked to their old choices. Hence finding the right balance between the two is challenging.

Benefits to Spotify:

This personalised playlist recommendation by Spotify is powered by data analytics that makes new playlists based on user preferences and listening history. It increases user retention and enhances user engagement and satisfaction as these tailored playlists create a unique listening experience. It gives Spotify a competitive advantage over its competitors as it encourages free users to opt for premium features to have a better experience.

2. Analysing Intermediary Role of Spotify in the Music Industry

Approach:

According to analysis done by (Prey et al., 2020) they examined how Spotify plays an intermediary role through monitoring their promotion patterns through their global Twitter account. For doing this analysis they collected between 2012 and 2018 from the Twitter account of Spotify. They studied Spotify's Twitter data over time from its promoted content focussed on albums, tracks, and playlists by categorising the content based on ownership by distinguishing them on the basis of major record labels and independent labels.

Results:

From the analysis it was found that Spotify tends to emphasise on major label artists limiting exposure for independent artists. It was found that the editorial team of Spotify that makes playlists like Rap Caviar and Today's Top Hits emphasize more on well-known artists as compared to emerging artists. This analysis reflects how advertising practices of Spotify exaggerates the existing inequalities in the music industry by decreasing the presence of independent artists.

Benefits to Spotify:

The Insights from this analysis can help Spotify to maintain the balance of exposure between major and independent labels to promote diversity. This kind of balanced playlist curation and giving opportunities to independent artists can strengthen Spotify's image as a neutral intermediary.

3. Promotion of Songs through Spotify Playlists:

Approach:

According to the research by (Aguiar & Waldfogel, 2021) they examined the impact of personalised In order to understand the role of Spotify playlists in song and artist success they took a discontinuity and instrumental approach where they analysed the outcomes of songs that were present in playlist to the songs that were not present in the playlist. They also did a regression discontinuity that helped them to understand the effects of playlist promotion by comparing the rating of tracks that just made it onto a playlist with those that narrowly missed the cut-off.

Results:

From this study it was found that the playlists played a significant role in the overall rating of the songs. Tracks that were present in these recommended playlists had more listeners and increased the visibility of artists. It was also found that Spotify holds considerable influence in creating trends in the music industry and promoting certain artists.

Benefits to Spotify:

From this study Spotify needs to implement transparent and inclusive promotion practices to prevent any potential regulatory challenges. Spotify also needs to ensure that they make fair promotion decisions and provide balanced exposure for both major and independent labels to maintain a healthy competition in the music streaming industry.

Analysis of Customer Rating and Review Dataset:

In this analysis I used the Spotify user ratings, reviews, and listening habits dataset from Kaggle and used python to do exploratory, quantitative and qualitative data analysis and understand user behaviour that could help Spotify to understand trends and enhance its features.

Ratings Analysis:

From the analysis of user ratings it was found that a majority of users around 35.9% rated Spotify positively with five stars while 12.7% users rated it as four stars and 11.2% users rated it as three stars. It was also found that a significant portion of users around 28.7% rated it one star and 11.6% rated it two stars indicating dissatisfaction among certain users.

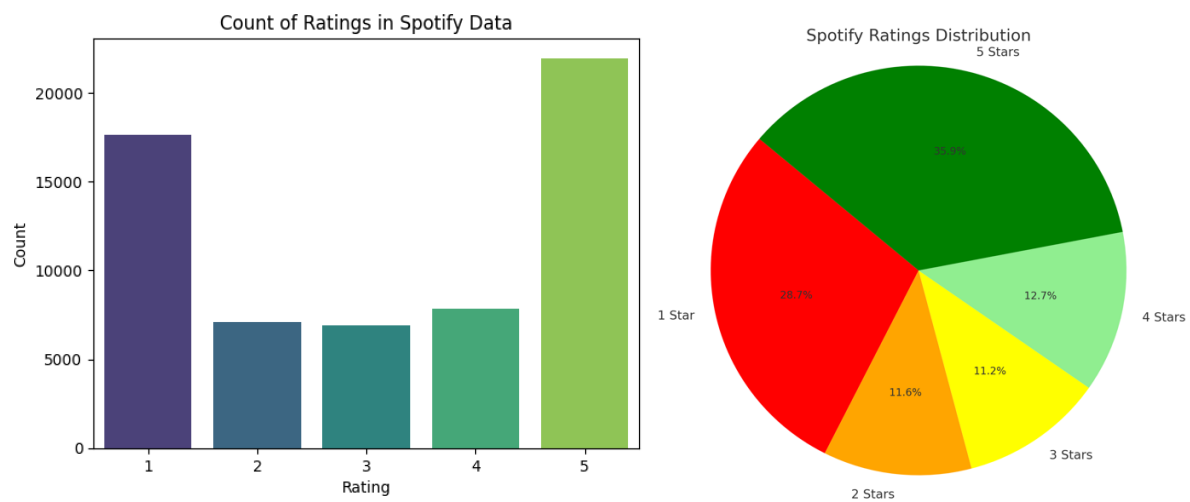


Figure 1: The above pie and bar chart reveals the rating given by users of Spotify for the services provided by the platform.

Sentiment Analysis on User Review:

From the analysis it was found that 65.8% of user review sentiment are positive whereas 23.2% are negative and 11% are neutral.

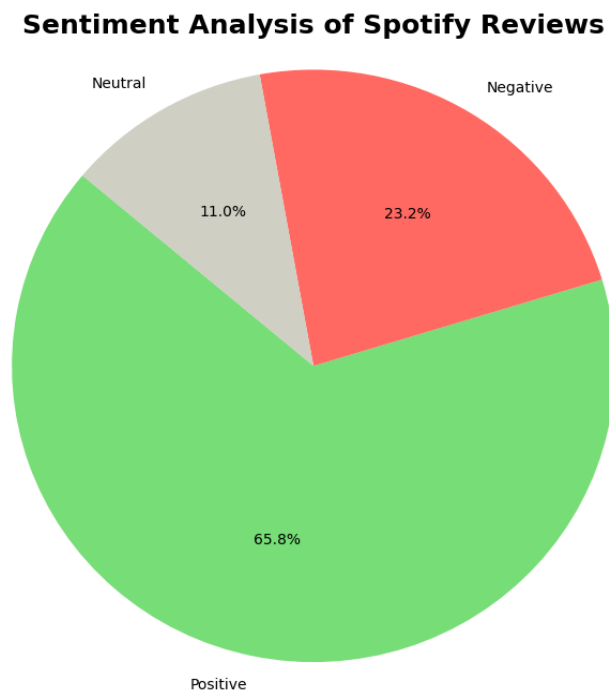


Figure 2: The pie chart represents sentiment analysis on the basis of reviews by Spotify users.

Sentiment and Thematic Analysis:

By using sentiment analysis and NLP, I identified common trends in user reviews. The word cloud below highlights the topics like song, playlist, app and podcast indicating they are more necessary for users than others. By doing a theme distribution analysis it was found that Music and App Performance are the most important themes as they account for 28.4% and 25.1% respectively. It was also found that Device Compatibility and Social Features are the least necessary themes accounting for 5% and 0.9% according to user reviews.

technology will ensure that the artists are paid fairly and transparently. Spotify can also use blockchain to simplify music listening by reducing the complexity of obtaining music rights (Öztürk et al., 2020).

2. AI generated music, Using AR and VR:

Spotify can use VR (Alonso et al., 2019) to organise live concerts in a virtual environment so that users can have unique concert experiences from their homes. Spotify can also develop AR applications so that their users can listen to music in innovative ways (Ludvigsen et al., 2010) such as modifying song tracks while listening or showing visuals related to tracks by using immersive graphics. Using AR and VR extensively will enhance user engagement which will help Spotify to make user focused decisions differentiating itself from other platforms. Spotify can also explore the unexplored domain of using AI to generate music, this will help them create unique content, revolutionise the entire music streaming industry and create new opportunity for emerging artists (Bozkurt et al., 2022).

3. Predictive Analytics, Security Enhancements and Smart Home Devices:

Spotify can expand its business scope by manufacturing and integrating with smart home devices like smart speakers and home entertainment systems. It can use IoT to understand the surrounding environment and provide music experiences tailored to activities or moods of its users (Alonso et al., 2019). Manufacturing smart home devices can help in making strategic decisions in terms of user engagement and market expansion. Spotify can leverage advanced predictive analytics to forecast future music trends that will help them in identifying upcoming music styles (Al-Jarro et al., 2022). By using predictive analytics Spotify can understand future trends and this will help them make future content related decisions. Spotify can also use quantum cryptography to ensure secure data transactions and safeguarding user data (Wang & Xie, 2022).

Conclusion:

Through this study it was found that Spotify uses data driven insights to improve user experience and decision making process. But still they have scope for improvement in certain areas where they can use emerging technologies to transform their platform into a global leader in its music streaming industry. Spotify has always leveraged the use of advanced technologies like ML and neural networks to personalize the user experience, solidify brand loyalty and increase user satisfaction. Still though this study it was found that many cognitive biases such as choice overload, loss aversion, algorithmic bias, popularity bias, etc that were affecting the decision making process and user experiences. By properly managing the cognitive biases with the help of data driven insights from user reviews and ratings can help Spotify to refine its platform strategy. Further Spotify needs to invest in emerging technologies like IoT, AI, blockchain and manufacturing smart devices. These strategies will help Spotify expand the scope of their business by targeting a larger audience and ensure sustainable growth in the rapidly evolving era of platformisation. Other platforms should use Spotify as a case study by enhancing their decision making process by balancing content promotion, making strategic changes based on user feedback, and investing in emerging technologies to manage competition and maintain diversity within the ecosystem.

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Appendix I

The Abbreviations:

Table 1 : Explanation or Full form of all the abbreviations used.

AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
VR	Virtual Reality
AR	Augmented Reality
RNN	Recurrent Neural Networks
CNN	Convolutional Neural Networks
SEM	Structural Equation Modelling
ANN	Artificial Neural Networks
BN	Bayesian Networks
IoT	Internet of Things
GBM	Gradient Boosting Machines

Appendix II

Practicality in applying emerging technologies:

1. Using the blockchain technology Spotify can make smart contracts which will trigger a payment to all the stakeholders at the end of each day on the basis of the number of times their work has been streamed. This will help artists from different regions to license their work easily.
2. The main issue faced by artists is that they are not fairly compensated due to complications in the present licensing system so Spotify can use blockchain to create a decentralized licensing platform so that artists can set their own licensing terms directly on the platform itself.
3. Spotify can use current advancements in AI to develop algorithms that will make personalised mood based music suggestions and real time song mashups that will give users a unique experience.

4. Using VR technology Spotify can create a virtual concert feature in their app where fans will not only listen to but also can interact with their favourite artists in real time. By introducing this Fans worldwide can attend virtual concerts and will help build a closer relationship between artists and fans.
5. Manufacturing devices like smart speakers and entertainment systems that integrate seamlessly with the Spotify streaming platform will massively increase the business scope of Spotify. By including features like voice commands and high quality sound will help Spotify expand in the market and have an advantage over its competitors like Amazon Alexa devices.
6. Advanced predictive is a powerful tool and will not only help Spotify determine the future market but also help them predict emerging artists and new innovative genres of music that have potential of reaching new heights.
7. This will help Spotify to make decisions regarding investing in upcoming artists and genres and staying ahead of market trends by attracting new users.

Appendix III

Analysis of Sentiment Score on the basis of user reviews:

On doing a sentiment analysis on the basis of user reviews it was found that sentiment scores are skewed towards positive sentiments making them approximately 60% of the reviews with maximum of them clustering around 0.0 (neutral) to 0.5 (positive) whereas negative sentiments comprises up of approximately 20% of the reviews clustering around -1.0 (negative) to 0.0 (neutral).

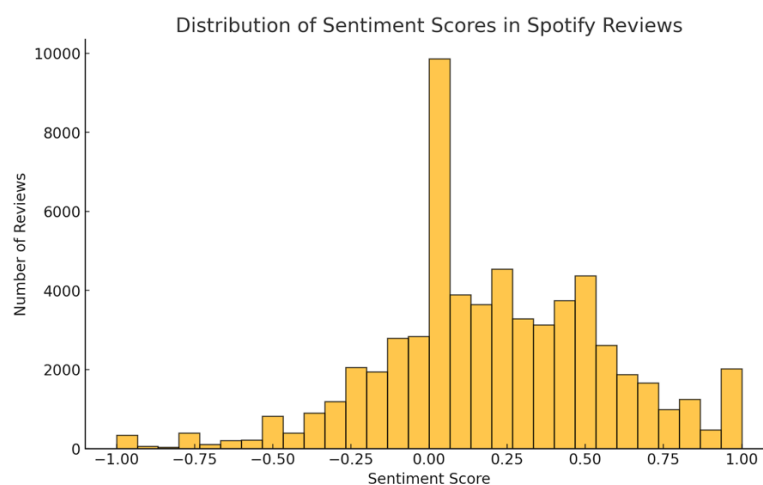


Figure 4: This above graph shows user sentiment on the basis of their reviews.

Appendix IV

Sentiment Analysis across Months:

The graph indicates significant fluctuations in sentiment scores over time. The graph indicates a steep rise between April and May with +53% points indicating a sudden increase in the overall user sentiment. Further there is a notable decline in user sentiments between March and April accounting for -27%.

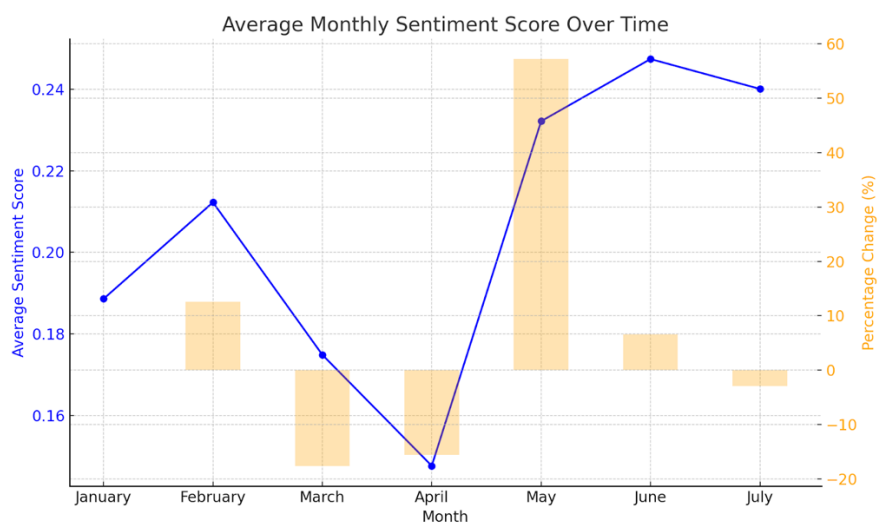


Figure 5: The above line charts represents user sentiment score across the months.

Appendix V

Understanding how Platform Involvement Increase Build Loyalty in Users

Approach:

According to a study by (Theadora et al., 2022) they used SEM and ANN techniques to understand the relationship between user involvement, engagement, and app loyalty. They collected data through an online questionnaire from Spotify users in Malaysia.

Results:

From the study it was found that user involvement played a huge role in increasing app trust. Through this method it was clear they users that are involved with Spotify are much more likely to stay loyal with the brand. It was also found that users who use Spotify tend to recommend it to others acting as free marketers for the platform.

Benefits to Spotify:

This study related to user involvement can help Spotify to make effective user retention strategies by increasing brand loyalty. Furthermore they should encourage cooperative activities between users and the platform such as exclusive content to promote deeper involvement. This will result in free word of mouth marketing with loyal users recommending Spotify to others.